

TANMAY AGRAWAL

SOFTWARE DEVELOPMENT ENGINEER | PYTHON BACKEND & AWS

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PROFESSIONAL SUMMARY

Python backend engineer building production APIs, event-driven AWS services and AI-enabled document workflows. Independently delivered multi-tenant webhook modernization, PostgreSQL performance fixes and client-data migration across AWS regions; worked directly with product, AI, operations and enterprise clients. Pibit.ai Top Performer of 2025.

TECHNICAL SKILLS

Languages & frameworks: Python, JavaScript, SQL; FastAPI, Flask, Node.js, Express, React, Celery, SQLAlchemy, Alembic
Cloud & DevOps: AWS Lambda, EC2, Fargate, ECR, S3, SQS, SES, API Gateway, CloudFormation, Parameter Store, CloudWatch, IoT Core; Docker, GitHub Actions
Databases & messaging: PostgreSQL, Amazon RDS, Redis, DynamoDB, Firebase Realtime Database, Firebase Cloud Messaging, MQTT, WebSockets
AI, testing & observability: Azure OpenAI (GPT-4o), Azure OCR, LLM integration, document extraction/classification, Pytest, LocalStack, Selenium, Sentry, Git
Engineering: REST APIs, webhooks, microservices, asynchronous processing, SQL optimization, caching, HLD/LLD, OAuth, JWT, API security

PROFESSIONAL EXPERIENCE

PIBIT.AI | Software Development Engineer (Backend) Jul 2025 - Present | Bengaluru

- Multi-tenant webhook modernization:** Independently redesigned a legacy enterprise webhook service with modular, client-isolated processing, validation and failure handling; implemented and reviewed the new architecture and monitored its first production week without observed defects.
- PostgreSQL migration & correctness:** Replaced inefficient UUID/text joins with native UUID schema and query changes, using EXPLAIN ANALYZE and Dalibo to identify bottlenecks; tested on production-like data and manually verified client workflows and billing figures before and after rollout.
- Production AI workflows:** Built and maintained Azure OpenAI GPT-4o document-extraction and classification integrations for insurance workflows, combining OCR outputs, client-specific processing and structured validation to support reliable downstream ingestion.
- Cross-region client migration:** Owned application refactoring and PostgreSQL/S3 data migration for clients moving from AWS ap-south-1 to us-east-1; removed region-specific hardcoding, preserved client identifiers and reconciled billing data, with no customer-visible downtime observed. An external team handled infrastructure relocation.
- PDF/OCR concurrency:** Reworked PDF rotation and Azure OCR processing with asyncio.gather and careful page-level resource management to avoid Lambda timeouts and memory exhaustion; reduced affected runs from over 15 minutes to approximately 1-2 minutes.
- Database & API performance:** Tuned slow APIs through query optimization and caching, observing one affected endpoint improve from roughly 10 seconds to 30-40 ms after successive changes; separated Amazon RDS read and write workloads to reduce contention.
- Redis-backed resilience:** Designed Redis caching for AWS Systems Manager Parameter Store to mitigate request throttling, with differentiated TTLs, concurrency safeguards and fallback to SSM if Redis is unavailable.
- Reporting-service foundation:** Built the FastAPI API and persistence foundation for an internal reporting service using PostgreSQL, Alembic, Redis job state and a Celery orchestration layer; other engineers implemented the core reporting logic.
- Notification platform:** Designed a Firebase Cloud Messaging notification service with user/device-token mappings, notification types, Firebase Realtime Database, multicast delivery and scheduled Celery workers.
- Audit trail & operational alerts:** Implemented queued, batched audit writes, entity-level history views and Slack notifications for configuration changes; separately integrated Sentry and CloudWatch with Slack alerts so US-hours operations staff could detect unexpected production failures.
- Enterprise APIs & client support:** Developed and maintained client-specific REST/webhook contracts, OAuth and signature-based integrations, validation rules and API Gateway controls; diagnosed production ingestion failures directly with customers, including resolving a complex issue during a live technical call.
- AWS automation & deployment:** Built an SES-S3-Lambda-SQS routing system in AWS CloudFormation to isolate automated testing from client-facing flows; containerized workloads with Docker/ECR, deployed Lambda/Fargate services and automated deployments for two Lambdas using GitHub Actions.
- Cross-functional engineering:** Partnered with operations on monthly billing reconciliation and complex SQL investigations, built client-specific reporting dashboards and translated product/client requirements into backend fixes; authored HLD/LLD and API contracts while mentoring four interns.

PIBIT.AI | Software Development Engineer Intern (Backend) Aug 2024 - Jun 2025 | Bengaluru

- **Large-scale ingestion automation:** Developed a Python/Selenium workflow on EC2 to process approximately 60,000-70,000 client records from S3 through an internal tool; used tmux for unattended runs, retained task IDs and skipped completed filenames when resuming failed batches.
- **Document AI & processing:** Implemented and supported production PDF/OCR, Azure OpenAI extraction/classification and asynchronous document-processing workflows; adapted parsers and data validation for varied enterprise-client formats.
- **Enterprise integrations & support:** Maintained AWS-backed client API/webhook integrations and Slack-enabled operational automation; investigated ingestion, document and delivery failures in production before continuing this work as a full-time engineer.

MITZWAH ENGG INC. INDIA | Full Stack Web Developer (Paid Project) Apr 2024 - Jun 2024 | Hybrid

- **National IoT management portal:** Independently built and deployed a paid React, Node.js and Express platform for the company's air-curtain fleet, enabling authenticated device management, telemetry views and location-based monitoring across India.
- **AWS IoT integration:** Implemented AWS IoT Core/MQTT publish-subscribe control and telemetry with DynamoDB storage, using device certificates provided by the hardware team; coordinated software-to-hardware integration without developing firmware.

EDUCATION

B.Tech, Computer Science and Engineering | ABES Engineering College, Ghaziabad | Aug 2025 | CGPA: 8.01/10

Class XII: 94.4% | Class X: 94.8% | Bal Vidya Mandir Senior Secondary School, Sambhal

AWARDS & ACHIEVEMENTS

Top Performer of 2025 - Pibit.ai; recognized for customer-focused engineering delivery.

1st Place, ACM-ACPC - National-level programming competition at ABES Engineering College; 2,500+ algorithmic problems solved across coding platforms.